

1 Affiliated Random Variables

Use the standard notation $\wedge$ for component-wise minimum and $\vee$ for component-wise maximum. Let the support $\mathcal{X}$ be a **sublattice**, which means that for all $x', x'' \in \mathcal{X}$, $x' \vee x'' \in \mathcal{X}$ and $x' \wedge x'' \in \mathcal{X}$, i.e. it is closed under component-wise min and max.

Definition 1.1. *The vector of random variables $X = (X_1, X_2, \dots, X_m) \in \mathcal{X}$ with joint density f are **affiliated** if for all $x', x'' \in \mathcal{X}$, we have*

$$f(x' \wedge x'')f(x' \vee x'') \geq f(x')f(x'').$$

For densities that are strictly positive in the interior of $\mathcal{X}$ and twice continuously differentiable, we just need to check for log-supermodularity over all cross-arguments, i.e.

$$\frac{\partial^2}{\partial x_i \partial x_j} \ln f \geq 0 \quad \forall i \neq j.$$

For example, consider the standard bivariate normal density function, which has log representation

$$\log f = \text{constant} - \frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1 - \rho^2)}$$

with the cross partial $\frac{\rho}{1-\rho^2} \geq 0 \iff \rho \geq 0$. So the standard bivariate normal density is affiliated iff the correlation is non-negative.